Independent reconstruction of regional climate-informed West Nile neuroinvasive disease forecasts

Reproducible research implementation for Raul Moreno

8 October 2026

Abstract

We implement the regional negative-binomial Bayesian regression and retrospective evaluation described by Harp et al. (2026), using independently acquired public county disease counts and gridded climate inputs. We reconstruct county and regional case summaries, climate correlations, probabilistic forecasts, and score uncertainty. The executed analysis is incomplete: NASA NLDAS soil moisture required by several published predictor sets returned HTTP 401. We therefore distinguish exact published predictor fits where inputs are available from an explicitly restricted-candidate analysis using PRISM and gridMET. Synthetic recovery, independent MCMC, predictive checks, and scoring identities test the implementation. No original author code, deposited data, submitted competition forecast, or digitized plot enters the analysis.

1 Scope and evidence

The target is Harp et al. (2026), GeoHealth 10(2), e2025GH001657. Written Sections 2.1–2.6, Tables 1–2, and Supporting Information S1 provide the specification. All plotted numbers in this report were calculated by this implementation from public inputs. No published image appears in the report. Reproduced means that we generated the specified quantity independently; it does not imply numerical identity with the publication. Input vintages and explicit interpretation choices remain material. Descriptive case maps and correlation maps are identified separately from model forecasts. The study-region locator is a schematic and is not a reproduced model result. Report figures carry an R prefix; source-paper figure identifiers are stated in their captions and in the panel registry. Independently reproduced panels: 1a, 1b, 1c, 1d, 2a, 2b, 2c, 2d, 3a, 3b, 3c, 3d, S13, S16, S17, S18, S20. Partially reproduced panels: 4, 5, S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S14, S15, S19, S21, S22, S23, S24. The competition comparisons and county mixed-model extension are not reproduced.

2 Inputs and spatial processing

CDC's public ArboNET dashboard supplies annual county WNND counts. We use its historical release finalized 16 September 2026 and subset 2005–2022. A separate CDC file supplies 2022 Connecticut counties, whose rows in the main file are unavailable planning-region placeholders. Missing county-year activity rows are interpreted as zero reported cases; explicit missing case values are rejected. Legacy Shannon/Oglala Lakota and Bedford identifiers are harmonized to Census 2020 boundaries. Alaska, Hawaii, and territories are excluded. County totals match independently downloaded contiguous-state totals in 17 of 18 years. Two Iowa cases in 2010 use unknown county code 19999; we retain this documented discrepancy rather than assign cases to a region without evidence.

Counties are assigned to NEON October 2024 polygons using Census 2020 population centers. Coastal centers outside a polygon use the nearest polygon in EPSG:5070; the exception log gives all distances. We use 2020 cartographic county polygons at 1:500,000. Each county climate mean weights grid cells by cosine latitude, including cells whose centers lie inside the original polygon. Counties without valid cell centers use the nearest valid cell at their population center. This approximates fractional-cell integration; display geometries alone are simplified. Domain maps are clipped to the continental viewing window to omit noncontinental fragments.

The paper's text selects regions with at least 1,000 historical cases, whereas Figure 1's caption describes a 50-case annual-mean threshold. Current inputs give Mid Atlantic 987 cases; we retain the 11 explicitly named regions to preserve the comparison. These 11 regions contain 92.7 percent of the independently assigned continental county cases, compared with the paper's statement of more than 93 percent. Regional climate weights are $w_c = \log H_c$, where H_c is each county's total 2005–2022 cases. Counties with $H_c = 0$ are excluded and $H_c = 1$ has zero weight. The regional covariate is $x_{rt} = \sum_{c \in r} w_c x_{ct} / \sum_{c \in r} w_c$.

We download 720 PRISM monthly grids from the NC State Climate Office's public THREDDS service, with Oregon State PRISM identified in their metadata. Variables are mean temperature, mean dew point (°C), precipitation (mm per month), and maximum vapor pressure deficit (hPa). The seasonal window is previous October–December and concurrent January–July. These contemporary reprocessed inputs need not equal the paper's 2023 input vintage. gridMET pentad PDSI is averaged by calendar month before county reduction; the accessible metadata describe a 1980–2019 reference period. The paper does not specify how pentads were reduced to months.

NLDAS soil moisture could not be downloaded anonymously (HTTP 401); its legacy GrADS endpoint returned HTTP 410. The paper does not identify the land-surface model or version. A proposed, unexecuted interpretation is Noah monthly v2.0, 0–100 mm layer moisture, whose midpoint is 50 mm. No alternate soil product is substituted here. Our explicit missing soil row limits the six-variable screening and any model requiring soil moisture. Data versions, URL hashes, access times, transformations, and failed attempts are recorded in the repository's provenance files. Large raw downloads are excluded from Git and downloadable by script.

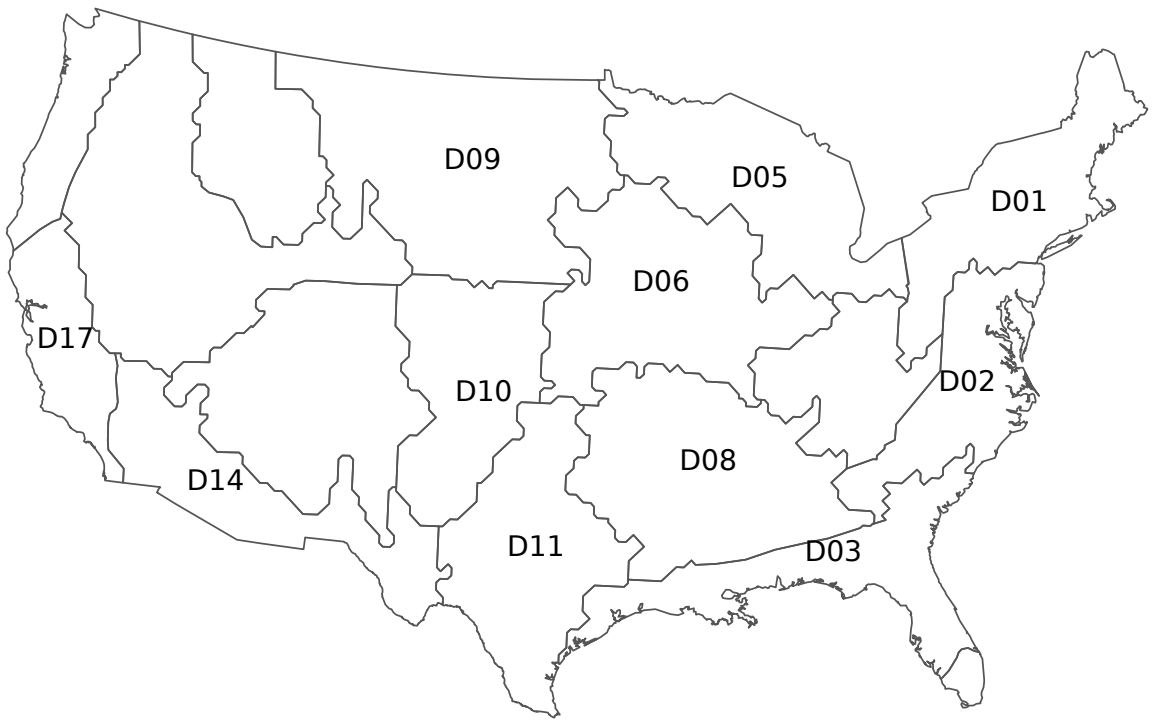


Figure R1: Study-region locator, redrawn from official NEON polygons. Labels identify the 11 modeled domains; unlabelled domains are excluded. This is a schematic, not a numerical reproduction.

3 Regression and retrospective evaluation

For each region and year, we model the raw count Y_t by

$$Y_t \mid \mu_t, \phi \sim \mathrm{NB2}(\mu_t, \phi), \qquad \log \mu_t = \alpha + \sum_j \beta_j z_{tj}, \qquad \mathrm{Var}(Y_t) = \mu_t + \mu_t^2/\phi. \tag{1}$$

Here z is the climate predictor centered and scaled by the training fold's mean and sample standard deviation. The univariable and bivariable models have one and two climate predictors. The historical benchmark has an intercept alone, with the same dispersion and prior treatment. All models are independent across regions. The supplement writes a linear expression for the mean but names `stan_glm.nb`, whose default link is logarithmic. We interpret the displayed linear expression as the linear predictor and use the log link. Counts are not logged before fitting the count likelihood.

We infer numerical prior settings from the official rstanarm defaults (Stan Development Team, 2026): independent $\alpha, \beta_j \sim N(0, 2.5^2)$ on the centered, standardized scale and $\phi \sim$ Exponential(1). These are weakly informative, not flat priors. The written supplement does not enumerate numerical prior scales. Our sampler includes the Jacobian for $\log \phi$. It uses a 7-degree-of-freedom multivariate Student proposal centered at the posterior mode, found with an analytic gradient, with scale 1.2 times the inverse finite-difference Hessian. Exact posterior-to-proposal weights correct the proposal. Thus a Gaussian curvature calculation does not replace the posterior. Four independent batches begin with 8,192 proposals each;

sample sizes double if importance effective sample size is below 4,000 or Pareto $k \geq 0.7$. Numerical pass also requires mean Monte Carlo error below 0.05 posterior standard deviations. This replaces the paper's four Stan chains of 500 warmup and 500 retained draws. Independent Metropolis chains test the numerical substitution.

Predictor selection ranks absolute Pearson correlations between $\log Y_t$ and the 60 potential month-variable combinations. Secondary variables must come from a different group: air conditions (temperature, dew point, VPD), antecedent soil conditions (PDSI, soil moisture), or precipitation. The executed restricted analysis screens 50 available candidates, then uses the same two-group constraint. We also fit the exact Table 1 predictor sets when their inputs are present. Screening and county climate weights use the full 18-year record, as described in the paper. They are not recomputed inside each validation fold; this can favor retrospective skill and precludes interpreting these scores as a prospective validation.

Each held-out year is predicted from a fit to the other 17 years. Thus early years are predicted using later observations. We generate 30,000 posterior predictive integer draws and save 23 empirical quantiles at 1, 2.5, 5, 10, 15,. . . ,95,97.5, and 99 percent. Seeds, parameter summaries, fold predictions, and diagnostics are saved. The full-model input pathway supports soil moisture when independent raw files become available, but those results are absent from this run.

3.1 Skill and uncertainty

We apply $\log(1+x)$ to observed counts and forecast quantiles. For $K = 11$ central intervals with lower and upper bounds l_k, u_k and miscoverage a_k, the score is

$$\mathrm{IS}_{a_k} = (u_k - l_k) + \frac{2}{a_k}(l_k - y)\mathbf{1}_{y<l_k} + \frac{2}{a_k}(y - u_k)\mathbf{1}_{y>u_k}, \tag{2}$$

$$\mathrm{WIS} = \frac{\frac{1}{2}|y - m| + \sum_{k=1}^{11} \frac{a_k}{2}\mathrm{IS}_{a_k}}{11.5}. \tag{3}$$

The median is m; the three components are width, overprediction, and underprediction (Bracher et al., 2021). We allocate the median term to the appropriate signed component, so the components sum exactly. Lower WIS is better. Percent improvement is $100(\mathrm{WIS}_B - \mathrm{WIS}_M)/\mathrm{WIS}_B$.

Regional uncertainty uses 10,000 paired resamples of 18 years. National scores weight regional scores by historical cases. National uncertainty resamples both regions and years and renormalizes those weights; this is an explicit interpretation of incompletely specified bootstrap details. Year draws are paired between models but independent between regions, so synchronous regional shocks are not preserved; this choice can affect national interval widths. The national score is not the score of a national total-count forecast. Intervals are bootstrap 2.5th and 97.5th percentiles. PIT uses the midpoint of the predictive CDF jump at an integer observation; interval coverage includes endpoints.

4 Numerical results and comparison

The restricted-candidate univariable model gives a national median improvement of 15.4% (4.6 to 25.5%), compared with the published 18.8% (4.7 to 27.7%). The difference is -3.4 percentage points. These estimates compare different predictor-selection spaces because soil moisture is missing.

The restricted-candidate bivariable model gives a national median improvement of 18.7% (5.6 to 30.3%), compared with the published 21.8% (7.0 to 30.7%). The difference is -3.1 percentage points. These estimates compare different predictor-selection spaces because soil moisture is missing.

As a benchmark sanity check, the independent regional mean historical-model WIS differs from the rounded published values in Supplementary Table S5 by a mean absolute 0.004 and a maximum absolute 0.010. The full regional comparison is saved in benchmark_comparison.csv. This check isolates the disease data and count-distribution model from the missing climate input.

Region	Paper uni (%)	Ours uni (%)	Paper bi (%)	Ours bi (%)
Pacific Southwest	25.3	24.6	20.8	17.6
Desert Southwest	11.3	10.7	7.2	3.0
Northern Plains	33.4	31.3	52.6	51.9
Central Plains	1.9	1.8	25.7	25.1
Southern Plains	32.9	10.0	31.6	19.0
Prairie Peninsula	29.5	22.5	33.5	21.9
Ozarks Complex	29.5	30.1	35.2	34.4
Great Lakes	10.7	12.4	10.5	8.3
Southeast	10.1	6.5	13.0	9.6
Mid Atlantic	17.6	17.8	32.3	35.2
Northeast	5.2	2.9	7.9	0.2
National	18.8	15.4	21.8	18.7

Table 1: Published Table 2 medians versus independent restricted-candidate screening. Published numbers are transcribed text references and do not enter fitting or generated plots. Full intervals and differences are saved in results/published_comparison.csv.

Region	Published predictor model	Paper (%)	Independent median [95% interval]
Pacific Southwest	uni	25.3	24.9 [-0.4, 42.2]
Desert Southwest	uni	11.3	9.9 [-9.9, 25.9]
Desert Southwest	bi	7.2	7.1 [-22.9, 27.0]
Northern Plains	uni	33.4	30.8 [10.0, 46.4]
Northern Plains	bi	52.6	52.0 [33.8, 63.3]
Central Plains	uni	1.9	1.9 [-12.8, 10.6]
Central Plains	bi	25.7	24.9 [-2.2, 42.1]
Ozarks Complex	uni	29.5	30.0 [2.3, 48.1]
Ozarks Complex	bi	35.2	34.7 [8.1, 54.4]
Mid Atlantic	uni	17.6	18.2 [-6.4, 35.0]
Mid Atlantic	bi	32.3	35.7 [8.2, 54.4]

Table 2: Separate fits using the exact written Table 1 predictor sets wherever available. Differences can still reflect data vintage, geometry, monthly aggregation, prior interpretation, and numerical integration.

Region	Paper primary	Selected primary	Paper secondary	Selected secondary
Pacific Southwest	tmean_03	tmean_03	soil_06	ppt_07
Desert Southwest	tmean_07	vpdmax_07	ppt_07	ppt_07
Northern Plains	tmean_05	tmean_05	pdsi_03	pdsi_03
Central Plains	tdmean_06	tdmean_06	pdsi_10	pdsi_10
Southern Plains	soil_10	ppt_03	ppt_03	pdsi_11
Prairie Peninsula	soil_03	pdsi_07	tmean_05	tmean_05
Ozarks Complex	pdsi_11	pdsi_11	tdmean_06	tdmean_06
Great Lakes	soil_03	tmean_01	tmean_01	ppt_12
Southeast	soil_01	vpdmax_05	vpdmax_05	ppt_10
Mid Atlantic	tmean_05	tmean_05	ppt_07	ppt_07
Northeast	soil_12	tdmean_10	tmean_05	ppt_12

Table 3: Predictor names encode variable and calendar month; months 10–12 are previous-year values. Selection uses all years and five available variables. Soil moisture candidates are absent, so equality with published selection is not expected.

Independent screening agrees with 5 of 11 published primary predictors and 6 of 11 secondary predictors. Both match in 4 regions: Northern Plains, Central Plains, Ozarks Complex, Mid Atlantic. In several soil-dependent regions, the available-input screen promotes the published secondary variable; this is a change in model specification rather than evidence that the original soil effect is absent.

Region	Table S1 slope	Independent slope	Independent bounds
Pacific Southwest	+0.271	+0.269	[+0.099, +0.439]
Desert Southwest	-0.928	-0.928	[-1.554, -0.302]
Northern Plains	-0.447	+0.552	[+0.242, +0.884]
Central Plains	-0.244	+0.153	[-0.110, +0.423]
Ozarks Complex	-0.237	-0.237	[-0.369, -0.108]
Mid Atlantic	+0.327	+0.334	[+0.125, +0.543]

Table 4: Univariable coefficients using the exact written predictors, in their raw input units. Each entry averages the posterior mean or 95 percent bound over the 18 held-out fits, matching the stated Table S1 summary. These averaged bounds are not a pooled posterior interval.

The Northern Plains temperature and Central Plains dew-point slopes have opposite signs to the entries printed in Supplementary Table S1. Other obtainable slopes agree closely in sign and magnitude. Our signs follow the independently calculated input associations and fitted likelihood. The written descriptions do not resolve whether the disagreement reflects a table error, an undocumented transformation, or different inputs. We retain the discrepancy; no sign was changed to match the reference. All fold-specific coefficients and dispersion estimates are saved in results/parameters.csv.

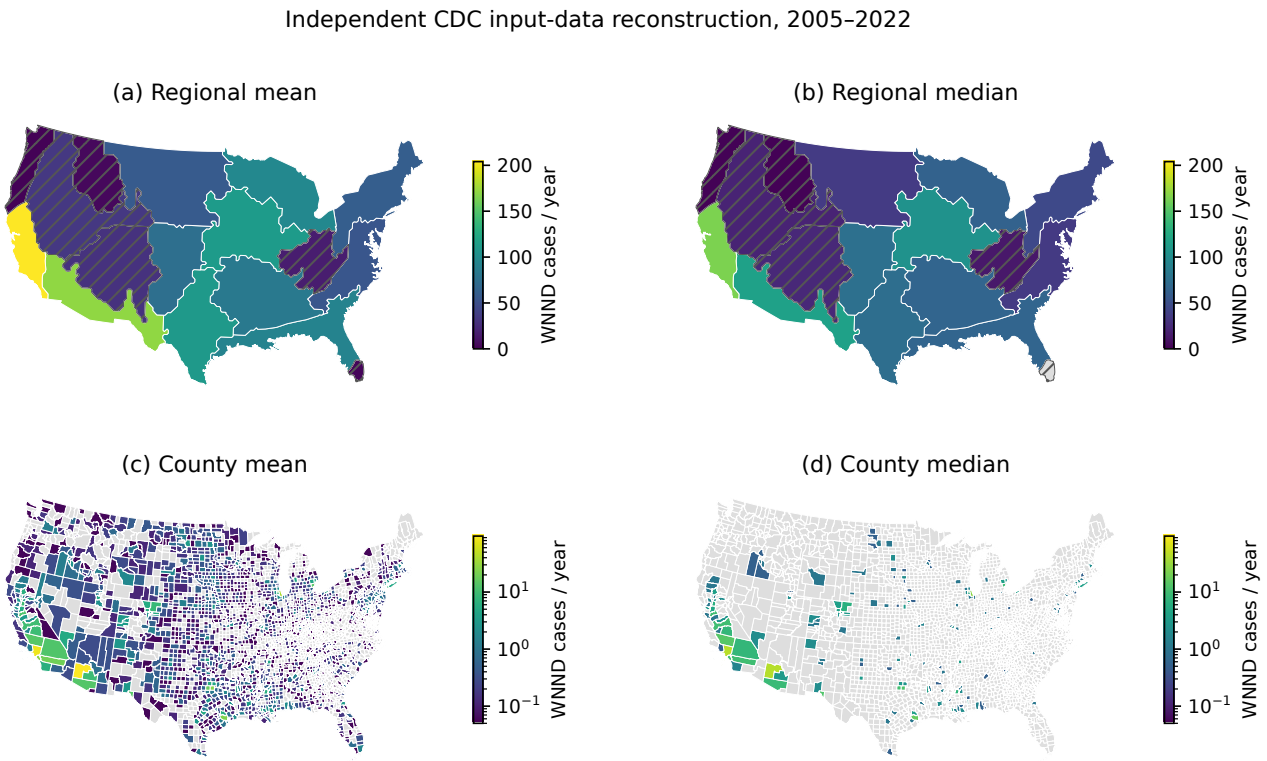


Figure R2: Independent Figure 1 reconstruction: (a,b) regional mean and median annual CDC WNND cases; (c,d) corresponding county quantities, 2005–2022. Grey county polygons have zero cases; hatching indicates excluded domains. Regional colors are linear and county colors logarithmic. These are descriptive observational summaries, not fitted model results.

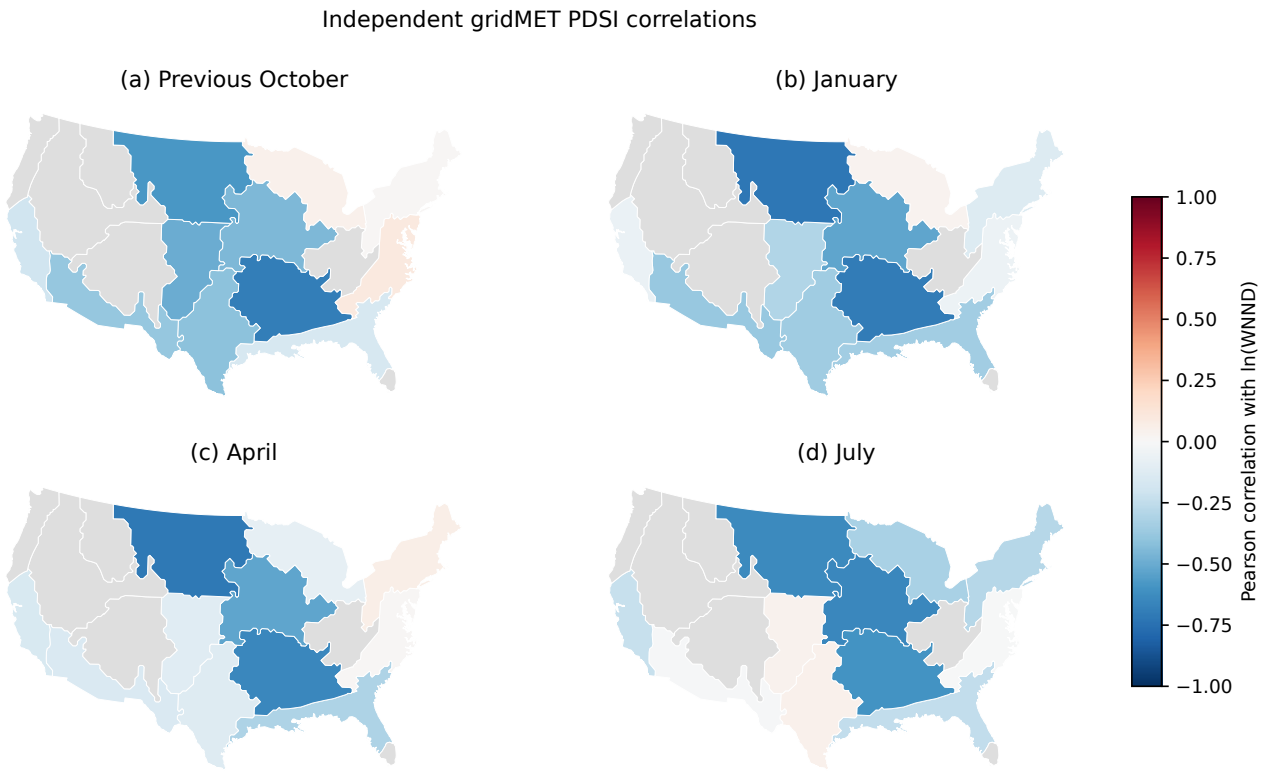


Figure R3: Independent Figure 2 reconstruction: correlation between log annual WNND and gridMET PDSI in (a) previous October, (b) January, (c) April, and (d) July. Positive PDSI denotes wetter conditions. Red means positive correlation and blue negative; grey domains are excluded. Each correlation uses 18 years and log-case county weights.

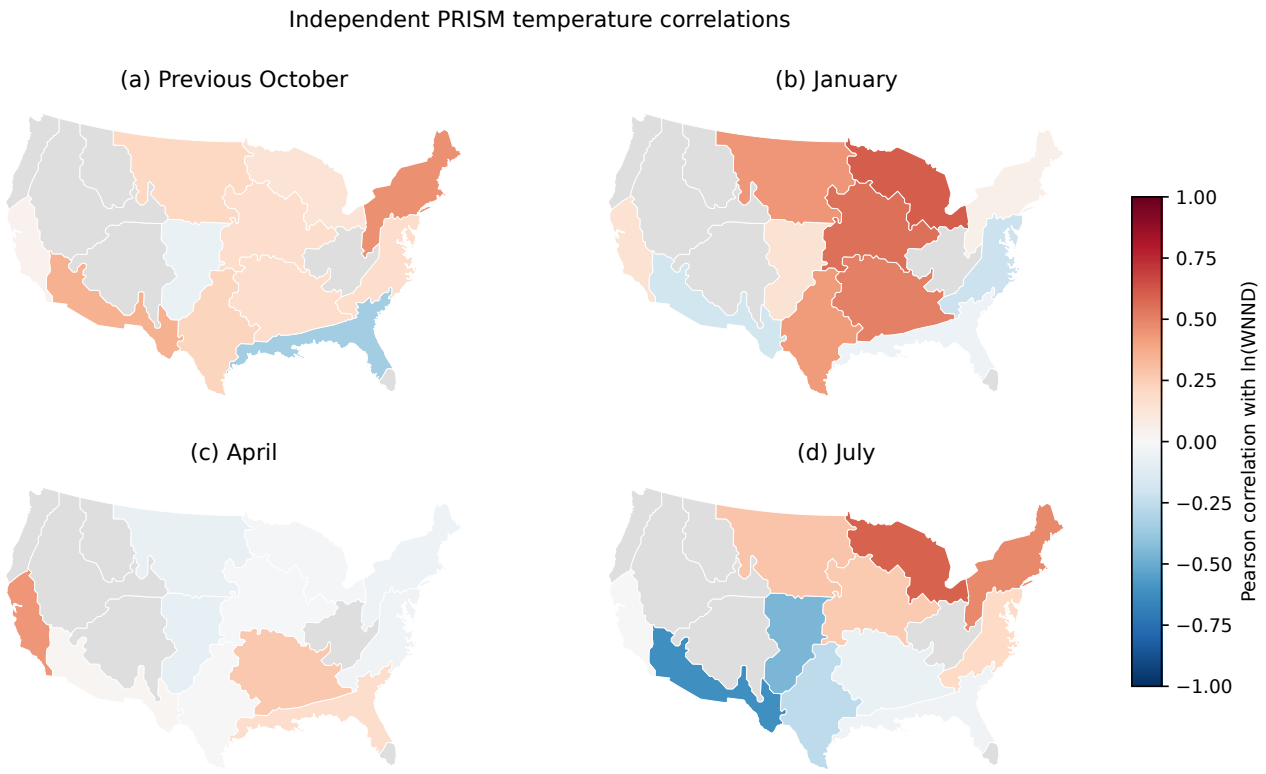


Figure R4: Independent Figure 3 reconstruction: the same calculation as Figure 2 for PRISM monthly mean temperature. The data are independently processed public grids, with no values extracted from published maps.

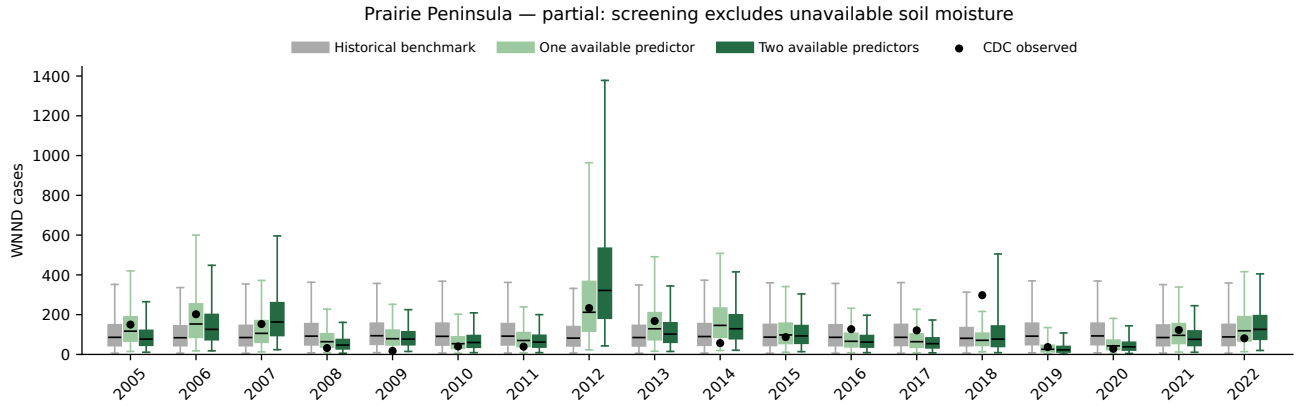


Figure R5: Partial Figure 4 reconstruction for Prairie Peninsula. Black points are CDC observations. Boxes show 25th, 50th and 75th predictive percentiles; whiskers span 2.5–97.5 percent. Grey is the historical benchmark. Green curves come from our restricted-candidate univariable and bivariable models because the published primary soil predictor is unavailable. They are independent calculations but do not complete reproduction of the published model.

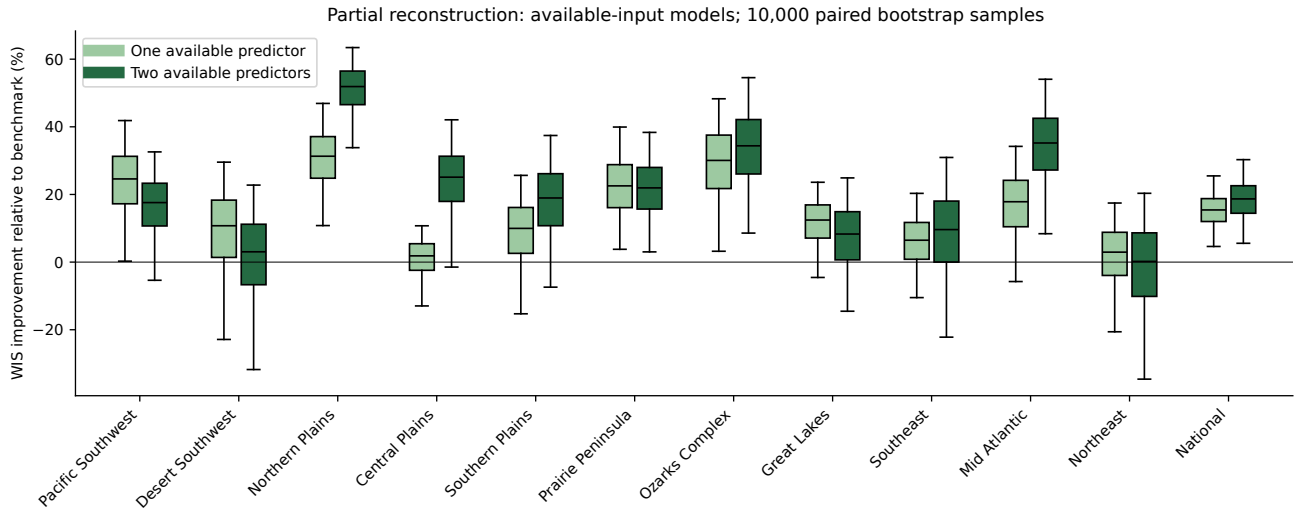


Figure R6: Partial Figure 5 reconstruction: relative WIS improvement for the independently selected available-input models. Boxes show bootstrap quartiles and whiskers span 95 percent intervals. National scores weight regional scores by case totals. The absent soil input limits comparability to the published full model.

4.1 Timing and magnitude of forecast errors

Region	Year	Observed	Forecast median	Signed error
Pacific Southwest	2014	533	232	-301
Desert Southwest	2021	1169	125	-1044
Northern Plains	2013	157	96	-61
Central Plains	2022	111	50	-61
Southern Plains	2012	721	215	-506
Prairie Peninsula	2018	298	77	-221
Ozarks Complex	2017	153	65	-88
Great Lakes	2012	328	86	-242
Southeast	2011	61	244	+183
Mid Atlantic	2012	134	55	-79
Northeast	2018	196	65	-131

Table 5: Largest absolute median forecast error in each region for the available-input bivariable model. Positive values mean overprediction. Units are WNND cases. These errors are calculated from saved held-out forecasts.

The publication identifies the Desert Southwest 2021 outbreak as an influential failure case. Our 2021 observation is 1169 cases; the available-input bivariable median is 125, with a 95% predictive interval [19, 505]. Exact published year-by-year predictions are not reported numerically in the text and were not digitized, so a numerical forecast-to-forecast error comparison is unavailable.

5 Validation and limitations

The likelihood agrees with an independent SciPy negative-binomial parameterization. The WIS calculation agrees with its 23-quantile pinball-loss identity; a point forecast equal to the observation scores exactly zero. Identical models remain identical under the paired bootstrap. Training-fold standardization is isolated from held-out covariates, and secondary predictor selection respects grouping. Across 792 leave-one-year-out fits, the smallest importance effective sample size is 20067, largest Pareto k is 0.457, and maximum mean Monte Carlo error is 0.018 posterior standard deviations. 0 fits fail the specified importance-sampling checks. The optimizer reports precision/status warnings for 0 proposals; these are retained in diagnostics. Since importance weights correct the proposal, these warnings alone do not establish a biased posterior, but poor weight diagnostics would fail the run. The known-parameter experiment uses 240 synthetic counts with intercept 3, slopes 0.55 and -0.4, and NB shape 4. All four true values fall inside their 95% posterior intervals. Over 24 repeated simulations, coverage is 91.7%, 100.0%, 100.0%, 100.0% for intercept, slope 1, slope 2, and shape. Four independent Metropolis chains give maximum R-hat 1.00, minimum bulk ESS 2680, and posterior mean differences below 0.027 posterior SD relative to importance sampling.

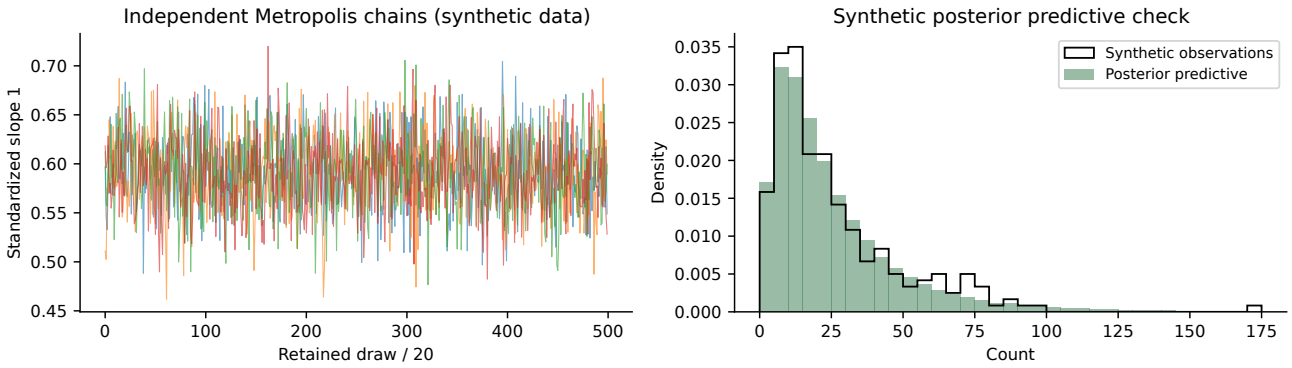


Figure R7: Implementation validation using synthetic data, separate from manuscript reproduction. Left: four independent Metropolis traces for a standardized slope. Right: observed synthetic counts and posterior predictive distribution. Known parameters and intervals are saved in synthetic_recovery.csv.

We also compare four Metropolis chains with importance sampling for the observed fold with largest Pareto k and for the Desert Southwest 2021 holdout using the published predictor pair. Across these two checks, maximum R-hat is 1.001, minimum bulk ESS is 2200, and the largest posterior mean difference is 0.026 posterior SD. All chain draws and check summaries are saved. Full-data posterior predictive checks compare mean, variance, and maximum counts for each of three available-input models in all 11 regions. 0 of 99 observed summaries lie outside their 95% posterior predictive intervals. These checks diagnose gross distributional mismatch; they are not independent tests of predictive skill. Held-out coverage, PIT values, and signed errors are saved separately. The principal limitation is missing NLDAS soil moisture. Additional discrepancies may arise from CDC revisions, spatial vintages, coastal assignments, grid-cell approximation, PDSI monthly averaging, and inferred defaults. Full-record predictor selection and climate weights allow information from held-out outcomes to enter the workflow, and are therefore a limitation of the retrospective design itself. No national result from the restricted-candidate run establishes or refutes the paper's full-model improvement. The climate associations are predictive regressions and do not isolate causal biological mechanisms.

The regional and county-level comparisons against the 2022 Forecasting Challenge are not reproduced because external forecast outputs are excluded by the project instructions. The county random-intercept

extension is not implemented in this regional pipeline. The main text's reference to logged county counts conflicts with its supplement's count likelihood and also needs an explicit resolution before that extension can be claimed reproduced.

6 Build procedure and outcome

The environment uses project-local Python 3.11.9 created by `/usr/bin/python3.11 -m venv .venv`. All Python commands use the explicit interpreter path. Exact installed dependencies are pinned in requirements.txt. The clean procedure is:

```
bash scripts/setup.sh
bash scripts/rebuild.sh
```

The first command recreates only .venv. The second downloads raw public inputs, reconstructs processed data, fits models, validates numerical methods, creates all plots, and compiles this report. For this delivered run, input downloads, processing, fitting, validation, plotting, and PDF compilation were executed; build logs and the final build outcome are stored in results. Cached raw files make repeat builds faster. Fold caches are checked against a hash of inputs, predictor specifications, random seeds, and modeling source before reuse. They are not independent input data. The repository includes saved processed inputs and computed outputs, so figures can also be regenerated offline.

TeX Live 2018 packages were checked with kpsewhich before writing the report: article, geometry, amsmath, graphicx, booktabs, longtable, float, xcolor, caption, subcaption, natbib, and hyperref. All were present. No system package was installed. The unavailable siunitx package is not used; units are typeset with standard mathematical roman text. The requested minimal-research and academic writing guidance is linked and summarized in docs/guidance.md; no project-specific vdv-damage content was used.

7 Figure-by-figure record

The machine-readable figure_registry.csv contains one row per panel, source documentation, input data, parameters, data command, plot command, and status. For compactness, command keys below mean:

```
G: .venv/bin/python src/prepare.py geography
C: .venv/bin/python src/prepare.py climate
F: .venv/bin/python src/forecast.py
P: .venv/bin/python src/plot.py
```

All numerical plots use saved independent output. R means independently reproduced, P means partially reproduced, and N means not reproduced. R does not assert exact agreement. Full source-specific assumptions are given above and in docs/method.md.

Panel	Quantity	Documentation / inputs / parameters	Status; data / plot command
1a	regional annual WNND mean	Section 2.2; Figure 1 caption; CDC; Census 2020; NEON Oct 2024; 2005-2022; zeros included	R; G / P; Descriptive observed-data reconstruction, not a model result; revised inputs and geography differ
1b	regional annual WNND median	Section 2.2; Figure 1 caption; CDC; Census 2020; NEON Oct 2024; 2005-2022; zeros included	R; G / P; Descriptive observed-data reconstruction, not a model result; revised inputs and geography differ
1c	county annual WNND mean	Section 2.2; Figure 1 caption; CDC; Census 2020; NEON Oct 2024; 2005-2022; zeros included	R; G / P; Descriptive observed-data reconstruction, not a model result; revised inputs and geography differ
1d	county annual WNND median	Section 2.2; Figure 1 caption; CDC; Census 2020; NEON Oct 2024; 2005-2022; zeros included	R; G / P; Descriptive observed-data reconstruction, not a model result; revised inputs and geography differ
2a	Pearson r: ln(WNND) vs previous October PDSI	Section 2.3; Figure 2 caption; CDC + gridMET; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
2b	Pearson r: ln(WNND) vs January PDSI	Section 2.3; Figure 2 caption; CDC + gridMET; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
2c	Pearson r: ln(WNND) vs April PDSI	Section 2.3; Figure 2 caption; CDC + gridMET; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
2d	Pearson r: ln(WNND) vs July PDSI	Section 2.3; Figure 2 caption; CDC + gridMET; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
3a	Pearson r: ln(WNND) vs previous October temperature	Section 2.3; Figure 3 caption; CDC + PRISM; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing

Panel	Quantity	Documentation / inputs / parameters	Status; data / plot command
3b	Pearson r: ln(WNND) vs January temperature	Section 2.3; Figure 3 caption; CDC + PRISM; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
3c	Pearson r: ln(WNND) vs April temperature	Section 2.3; Figure 3 caption; CDC + PRISM; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
3d	Pearson r: ln(WNND) vs July temperature	Section 2.3; Figure 3 caption; CDC + PRISM; 18 years; log-total-case county weights	R; C, F / P; Independent correlations; numerical differences expected from input vintage and spatial processing
4	Prairie Peninsula annual probabilistic forecasts	Sections 2.4-2.5; Table 1; Text S2; CDC + available PRISM/gridMET; 17 training years; 23 quantiles; NB2	P; F / P; Published primary soil moisture unavailable; generated restricted-candidate forecasts and benchmark
5	Regional and national percent WIS improvement	Section 2.5; Text S1; Figure 5; Saved independent forecasts; Log1p WIS; 10000 paired bootstrap samples	P; F / P; National and regional restricted-candidate analysis omits soil moisture; published-pair scores saved where possible
S1	Northeast monthly climate correlations	Section 2.3; Figure S1 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S2	Mid Atlantic monthly climate correlations	Section 2.3; Figure S2 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S3	Southeast monthly climate correlations	Section 2.3; Figure S3 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S4	Great Lakes monthly climate correlations	Section 2.3; Figure S4 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing

Panel	Quantity	Documentation / inputs / parameters	Status; data / plot command
S5	Prairie Peninsula monthly climate correlations	Section 2.3; Figure S5 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S6	Ozarks Complex monthly climate correlations	Section 2.3; Figure S6 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S7	Northern Plains monthly climate correlations	Section 2.3; Figure S7 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S8	Central Plains monthly climate correlations	Section 2.3; Figure S8 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S9	Southern Plains monthly climate correlations	Section 2.3; Figure S9 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S10	Desert Southwest monthly climate correlations	Section 2.3; Figure S10 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S11	Pacific Southwest monthly climate correlations	Section 2.3; Figure S11 caption; CDC; PRISM; gridMET; soil row missing; Oct(-1)-Jul; mask abs(r)¡0.32	P; F / P; Five of six climate rows independently computed; soil moisture row explicitly missing
S12	Northeast annual probabilistic forecasts	Table 1; Text S2; Figure S12 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	P; F / P; Soil moisture required by published model unavailable; restricted-candidate curves and benchmark shown
S13	Mid Atlantic annual probabilistic forecasts	Table 1; Text S2; Figure S13 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	R; F / P; Both published predictor sets independently fitted; input/prior/numerical assumptions documented
S14	Southeast annual probabilistic forecasts	Table 1; Text S2; Figure S14 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	P; F / P; Soil moisture required by published model unavailable; restricted-candidate curves and benchmark shown

Panel	Quantity	Documentation / inputs / parameters	Status; data / plot command
S15	Great Lakes annual probabilistic forecasts	Table 1; Text S2; Figure S15 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	P; F / P; Soil moisture required by published model unavailable; restricted-candidate curves and benchmark shown
S16	Ozarks Complex annual probabilistic forecasts	Table 1; Text S2; Figure S16 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	R; F / P; Both published predictor sets independently fitted; input/prior/numerical assumptions documented
S17	Northern Plains annual probabilistic forecasts	Table 1; Text S2; Figure S17 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	R; F / P; Both published predictor sets independently fitted; input/prior/numerical assumptions documented
S18	Central Plains annual probabilistic forecasts	Table 1; Text S2; Figure S18 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	R; F / P; Both published predictor sets independently fitted; input/prior/numerical assumptions documented
S19	Southern Plains annual probabilistic forecasts	Table 1; Text S2; Figure S19 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	P; F / P; Soil moisture required by published model unavailable; restricted-candidate curves and benchmark shown
S20	Desert Southwest annual probabilistic forecasts	Table 1; Text S2; Figure S20 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	R; F / P; Both published predictor sets independently fitted; input/prior/numerical assumptions documented
S21	Pacific Southwest annual probabilistic forecasts	Table 1; Text S2; Figure S21 caption; CDC; PRISM/gridMET; 18 LOYO forecasts; NB2; 23 quantiles	P; F / P; Soil moisture required by published model unavailable; restricted-candidate curves and benchmark shown
S22	Absolute WIS improvement	Section 2.5; Figure S22; Saved independent scores; 10000 paired bootstraps	P; F / P; Restricted-candidate model comparison
S23	univariable PIT calibration histogram	Figure S23; Text S2; Saved posterior predictive draws; 198 held-out predictions; 10 equal bins; midpoint PIT	P; F / P; Restricted-candidate models; discrete PIT convention assumed
S24	bivariable PIT calibration histogram	Figure S24; Text S2; Saved posterior predictive draws; 198 held-out predictions; 10 equal bins; midpoint PIT	P; F / P; Restricted-candidate models; discrete PIT convention assumed

Panel	Quantity	Documentation / inputs / parameters	Status; data / plot command
Table S3 / Section 3.2 competition comparisons	County mixed-model and regional ensemble forecast skill	Section 2.6; Text S3; Table S3; Competition forecast submissions excluded; Not evaluated	N; not run; County extension not run; competition outputs excluded by independence constraint

A Supplementary figure attempts

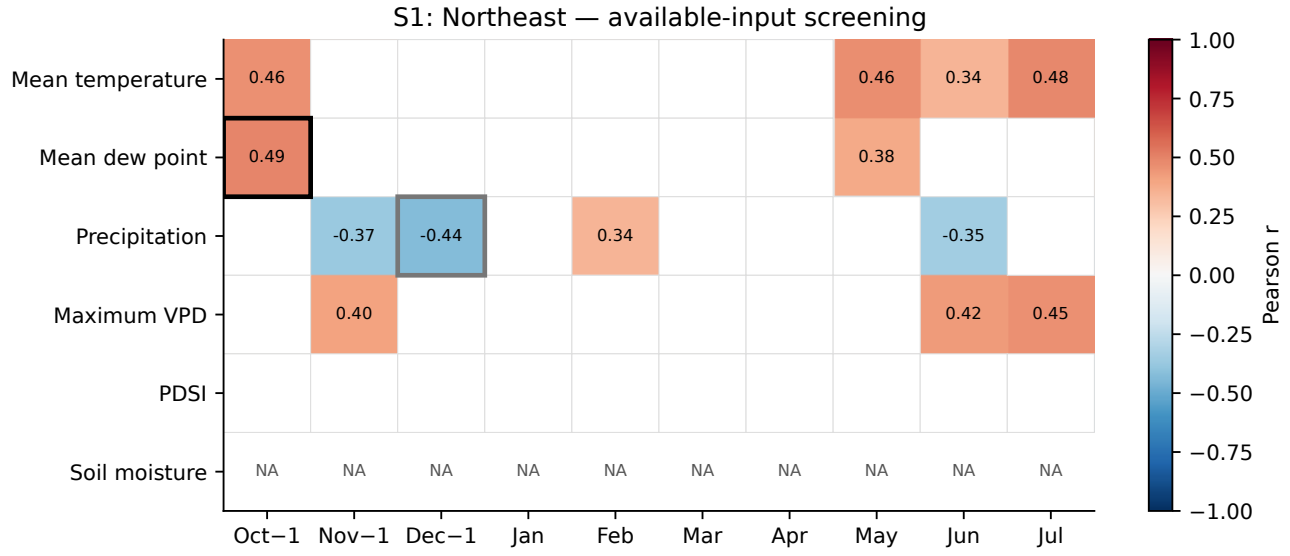


Figure R8: Figure S1: Northeast monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

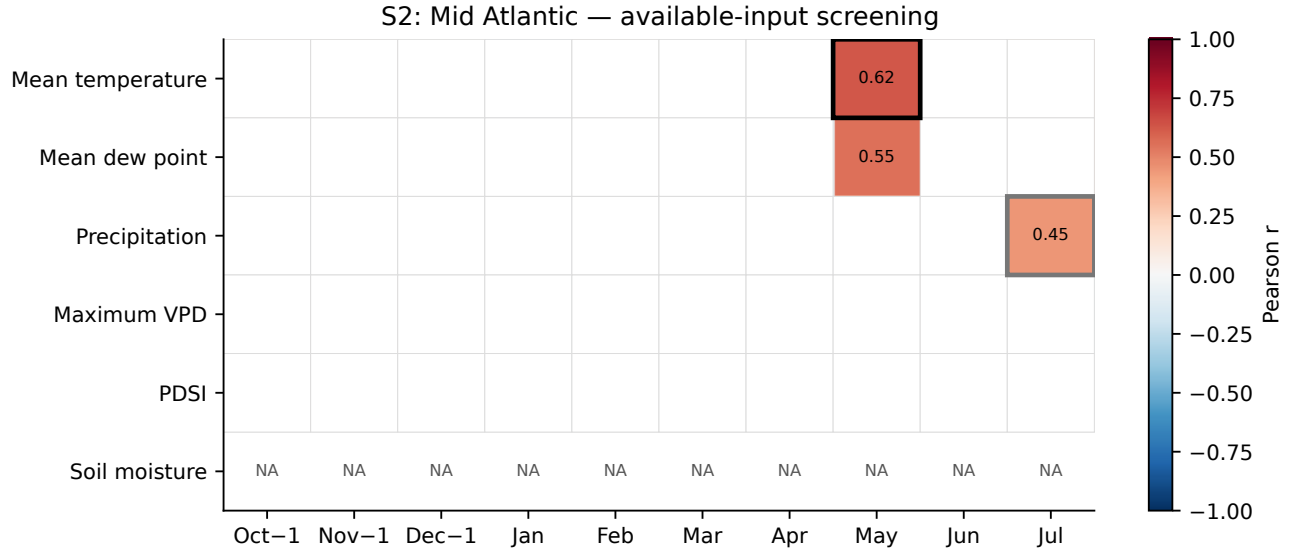


Figure R9: Figure S2: Mid Atlantic monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

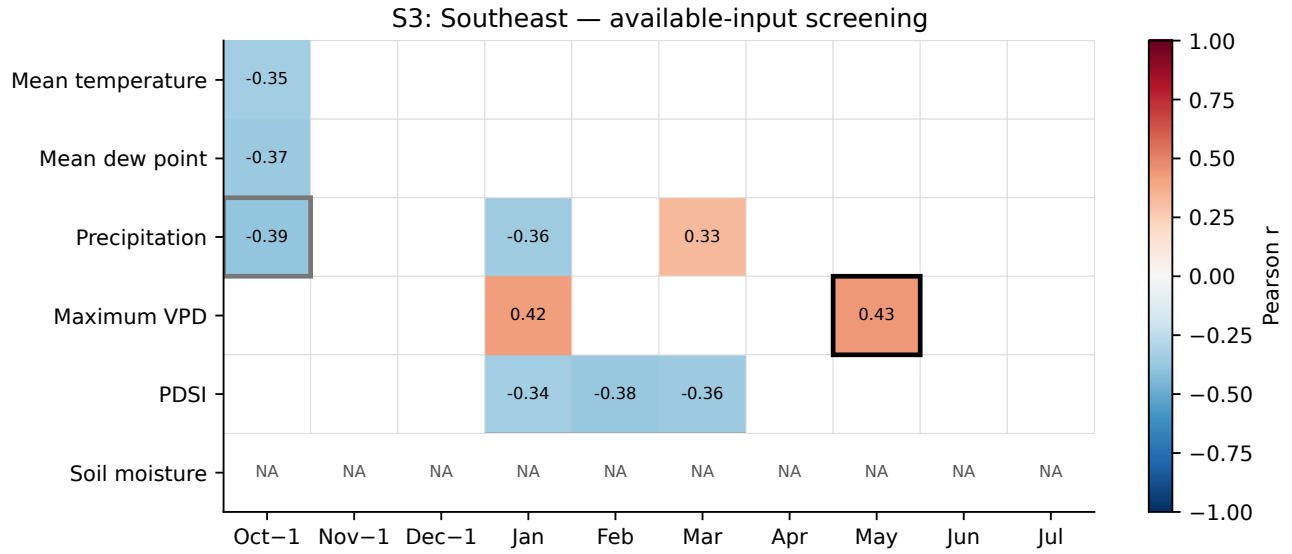


Figure R10: Figure S3: Southeast monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

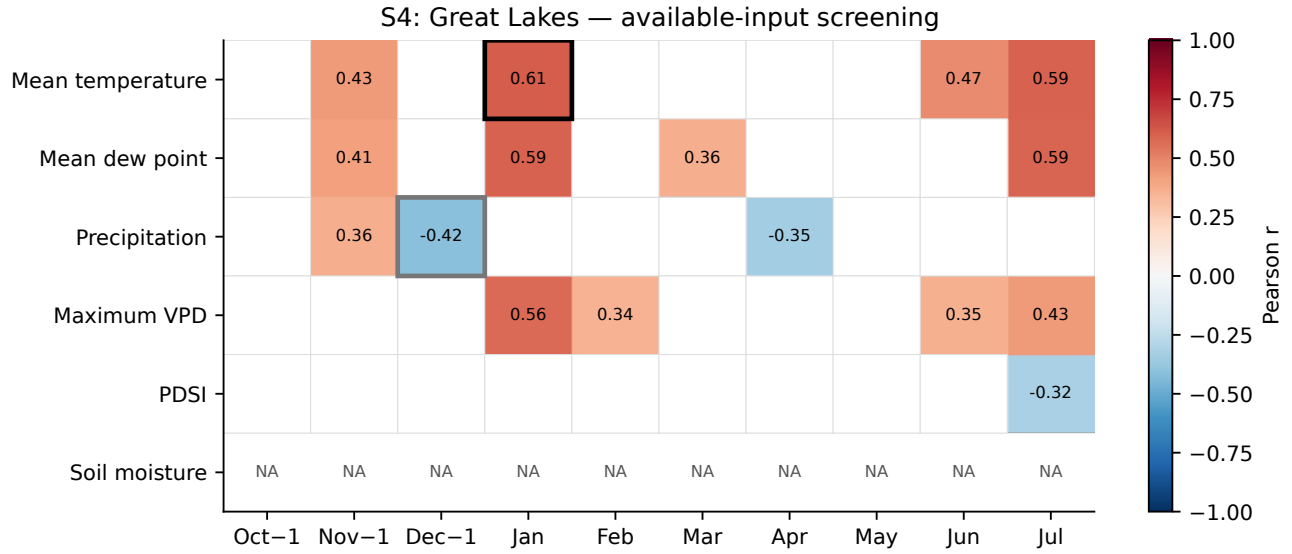


Figure R11: Figure S4: Great Lakes monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

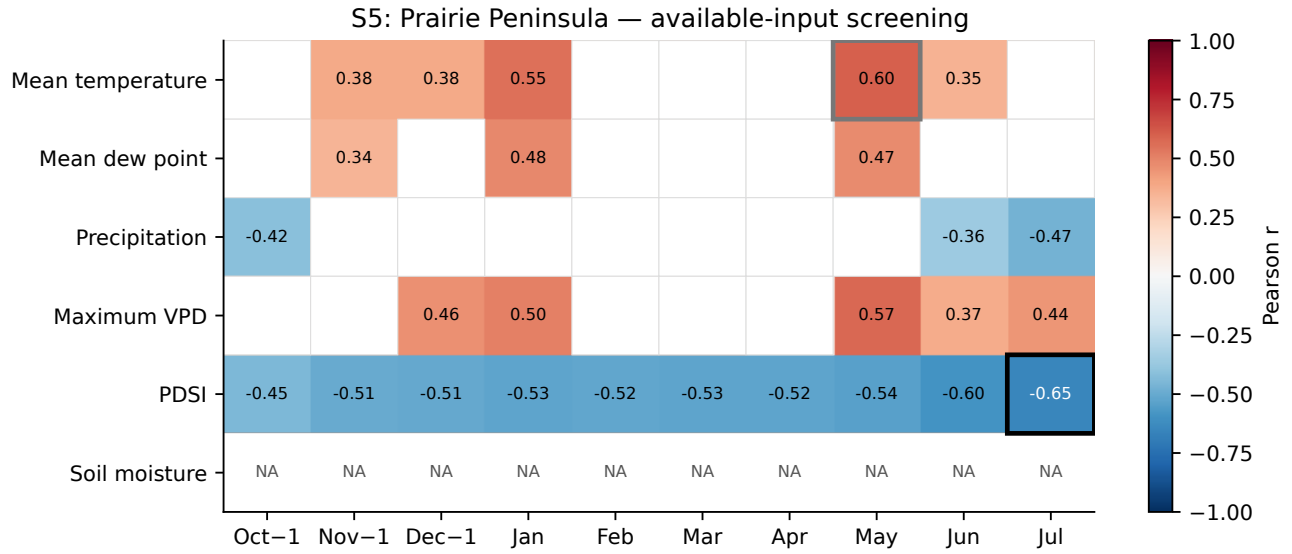


Figure R12: Figure S5: Prairie Peninsula monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

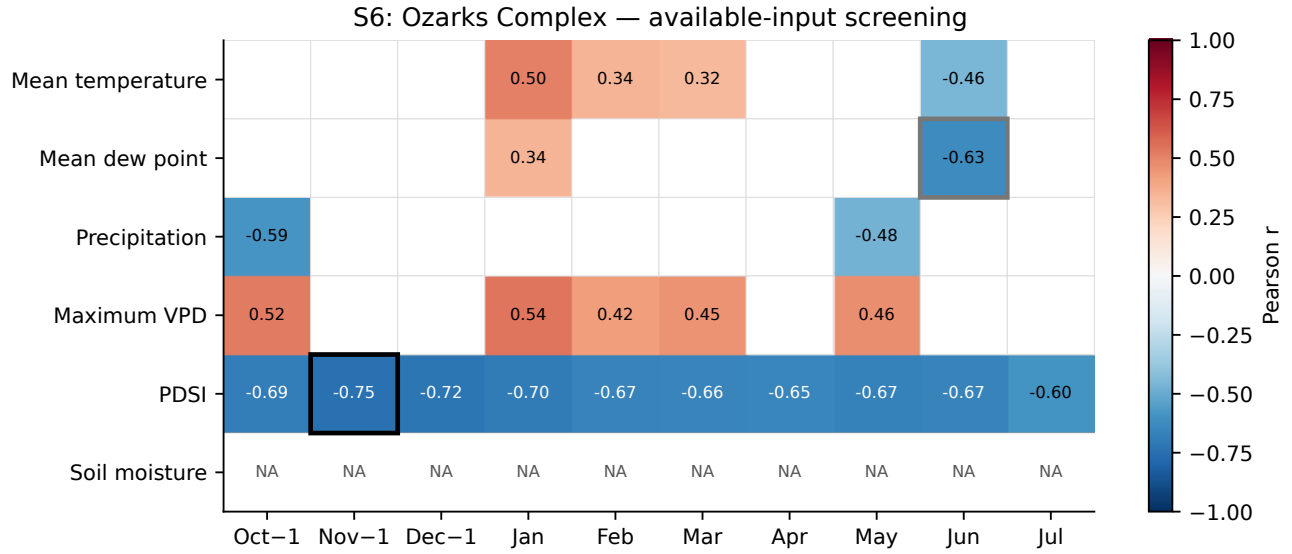


Figure R13: Figure S6: Ozarks Complex monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

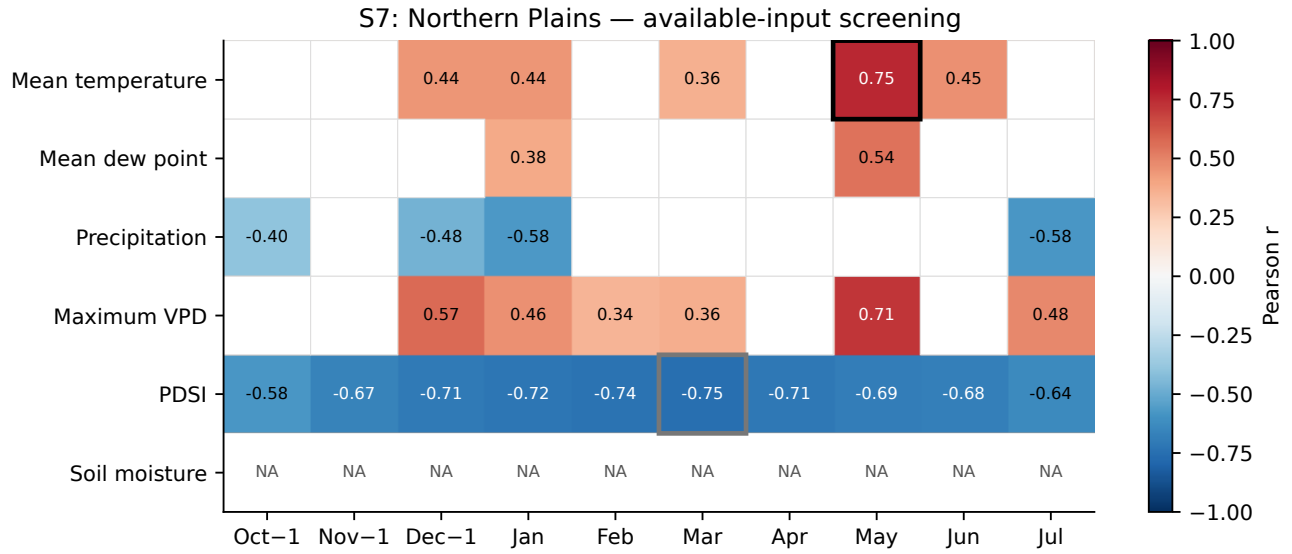


Figure R14: Figure S7: Northern Plains monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

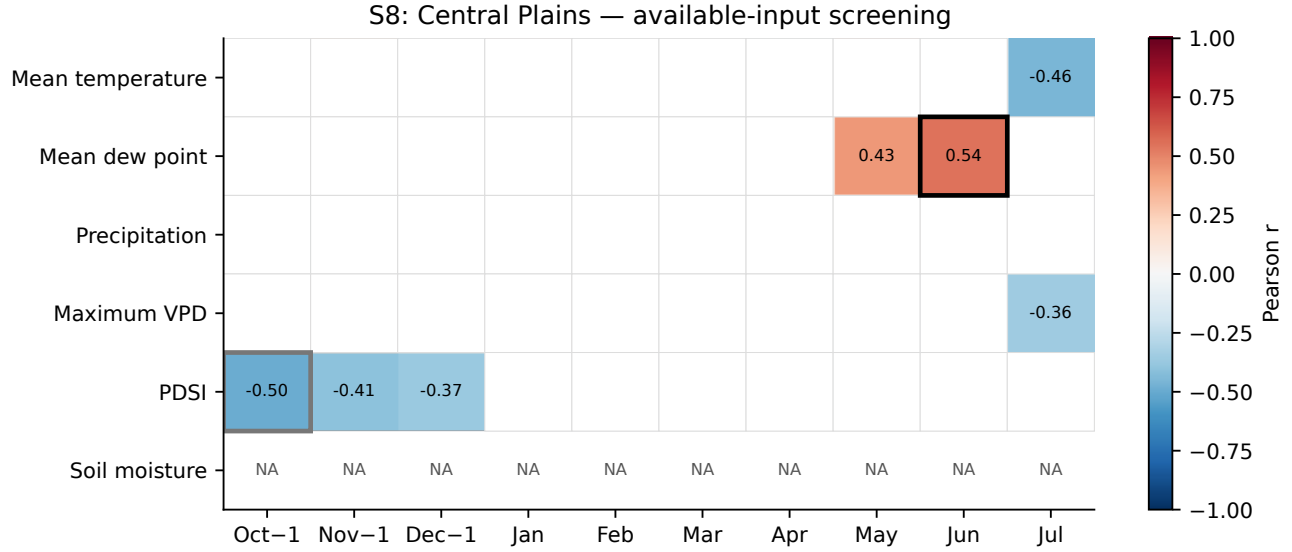


Figure R15: Figure S8: Central Plains monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

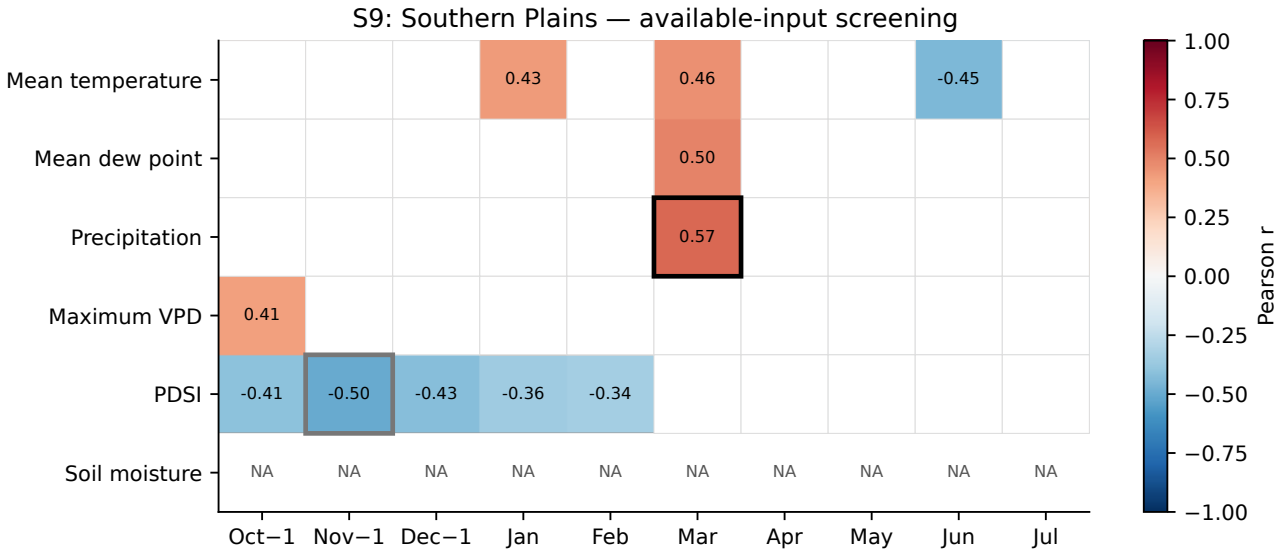


Figure R16: Figure S9: Southern Plains monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

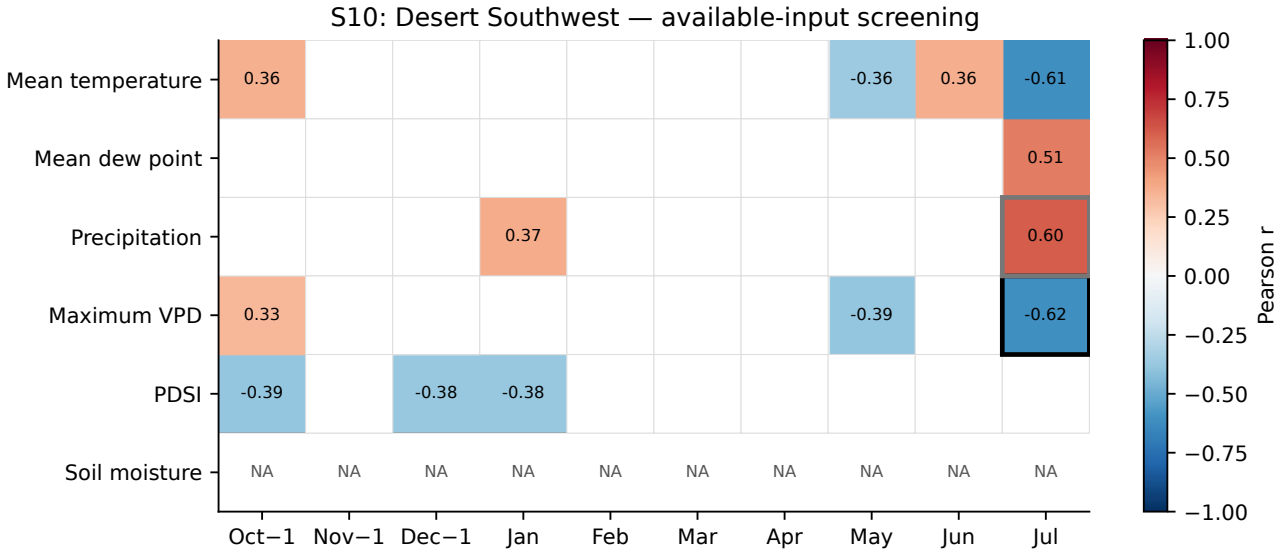


Figure R17: Figure S10: Desert Southwest monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

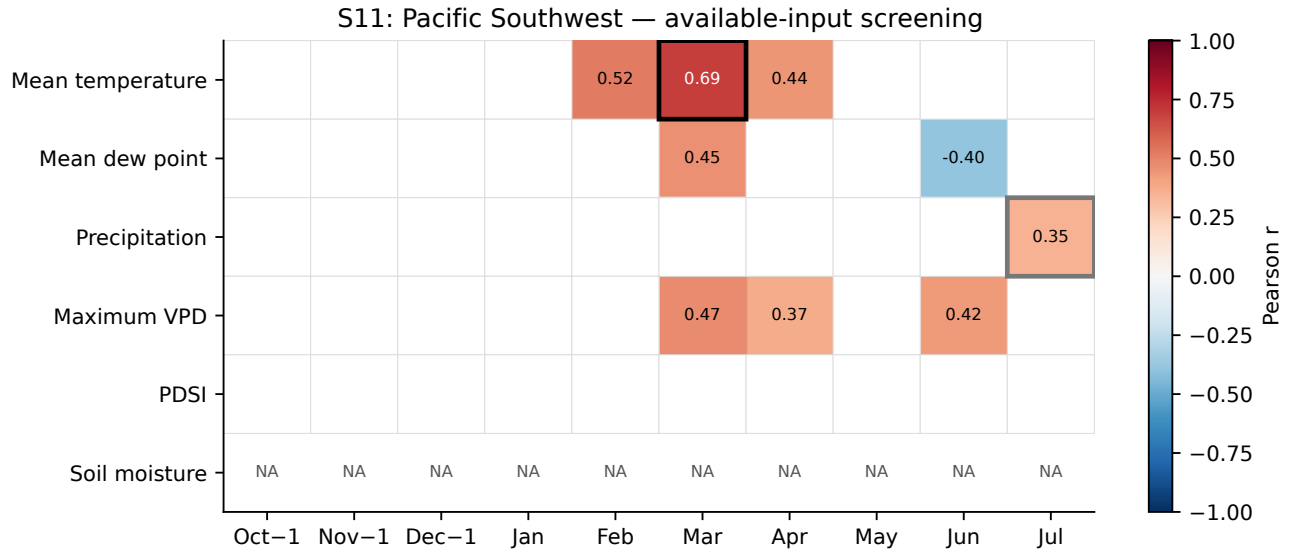


Figure R18: Figure S11: Pacific Southwest monthly climate correlations. partially reproduced. Correlations use 18 years; white cells mask absolute r below 0.32, NA denotes unavailable soil moisture, and black/grey borders mark selected primary/secondary available predictors.

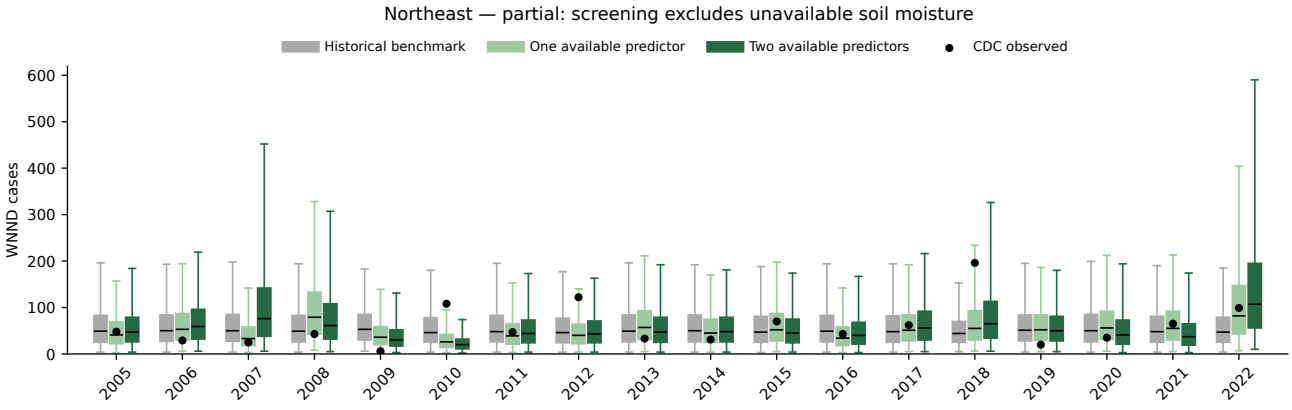


Figure R19: Figure S12: Northeast annual probabilistic forecasts. partially reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

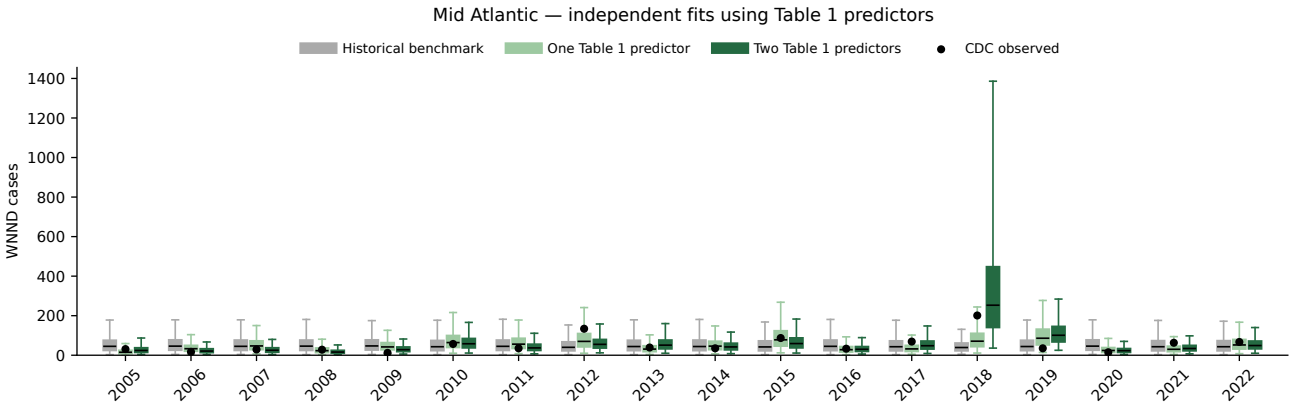


Figure R20: Figure S13: Mid Atlantic annual probabilistic forecasts. independently reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.



Figure R21: Figure S14: Southeast annual probabilistic forecasts. partially reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

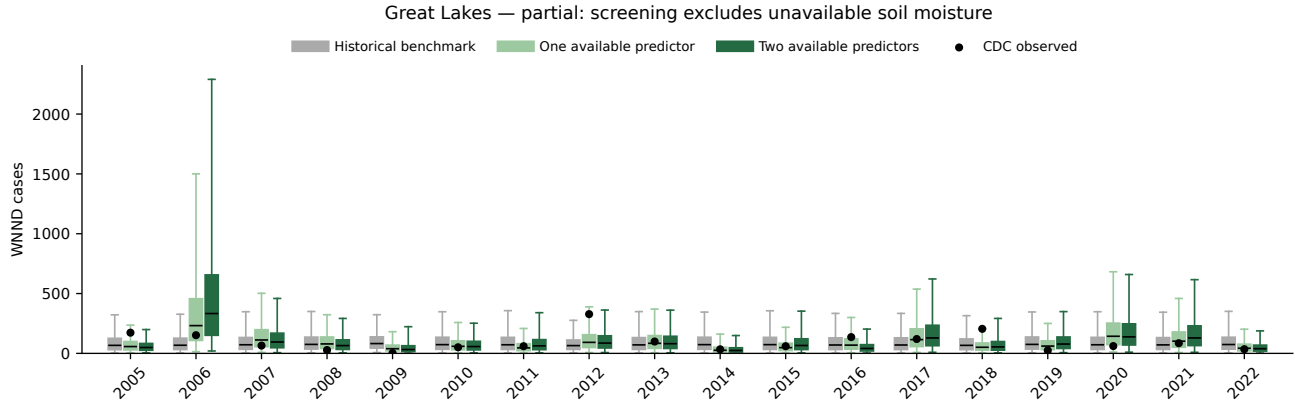


Figure R22: Figure S15: Great Lakes annual probabilistic forecasts. partially reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

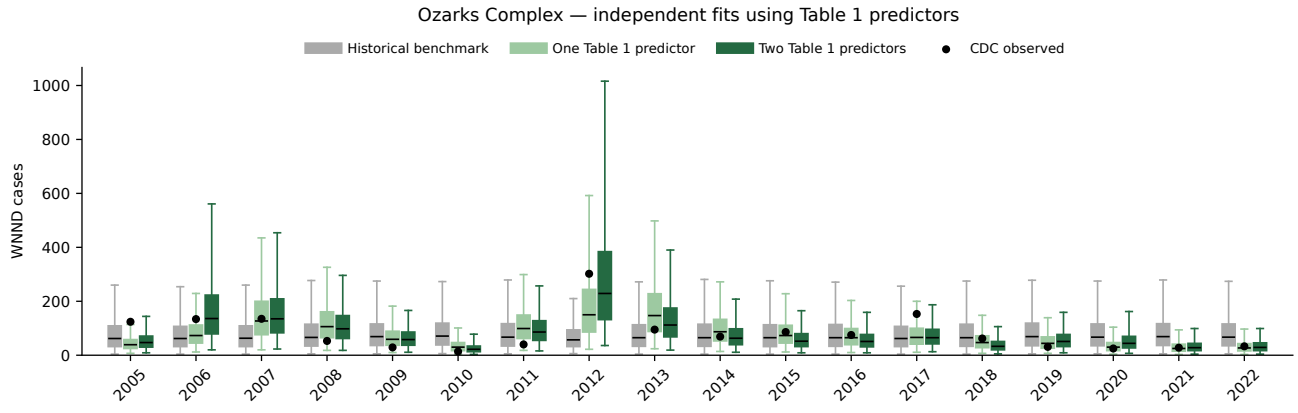


Figure R23: Figure S16: Ozarks Complex annual probabilistic forecasts. independently reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

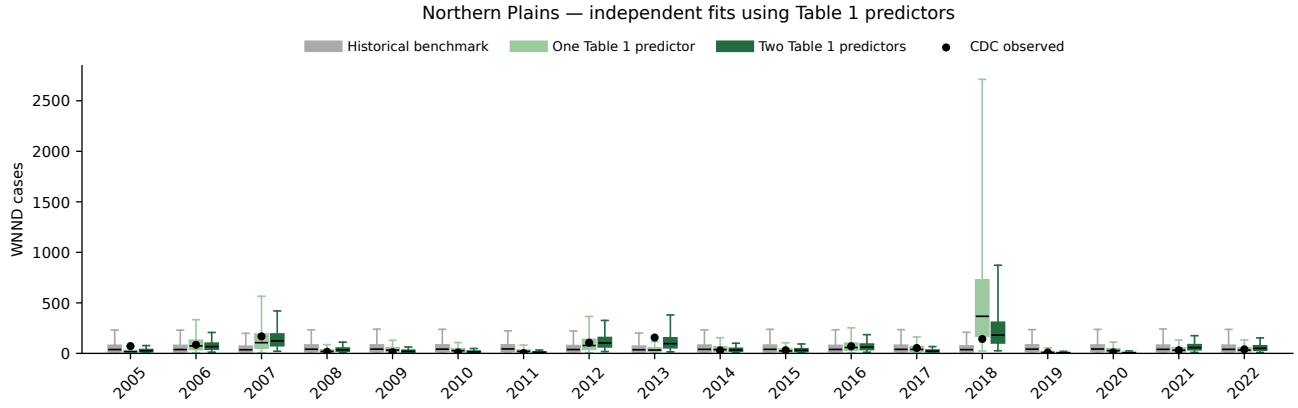


Figure R24: Figure S17: Northern Plains annual probabilistic forecasts. independently reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

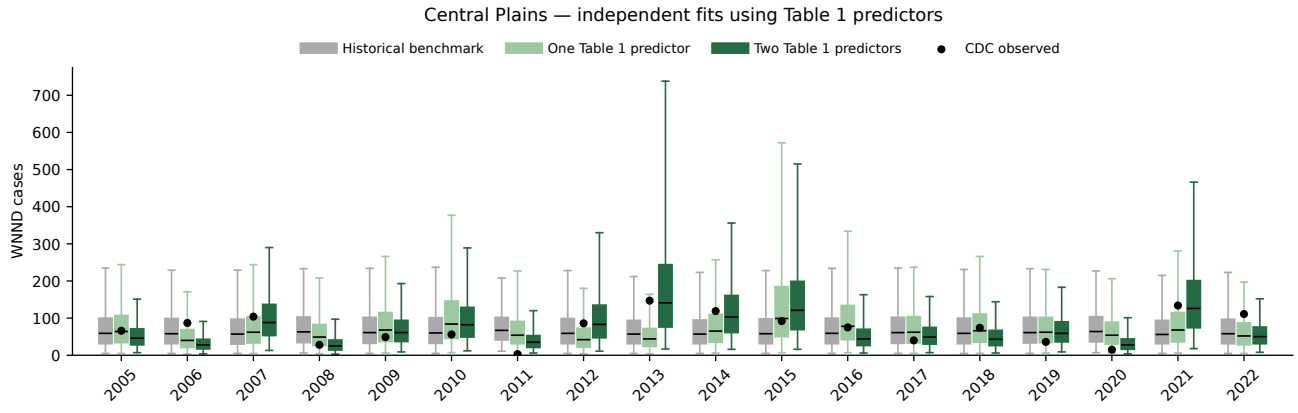


Figure R25: Figure S18: Central Plains annual probabilistic forecasts. independently reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

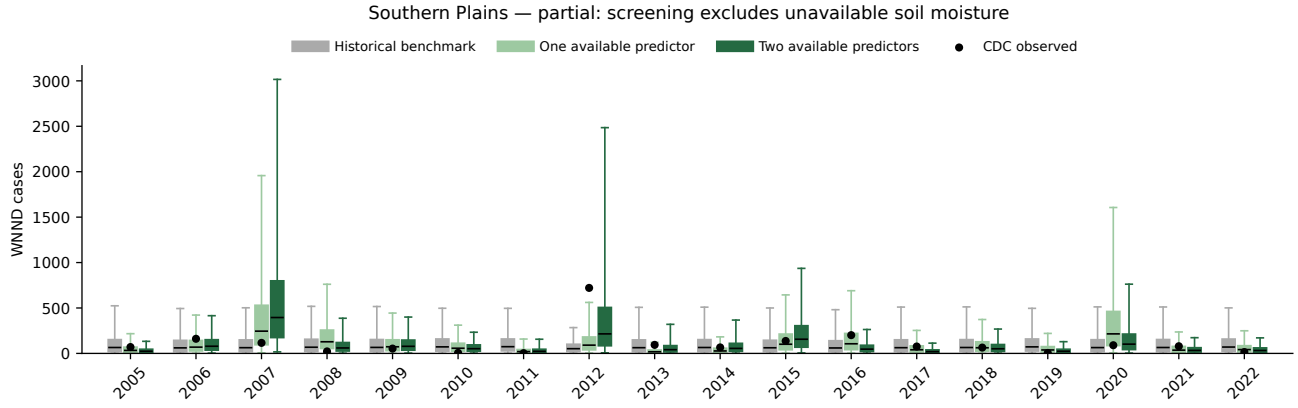


Figure R26: Figure S19: Southern Plains annual probabilistic forecasts. partially reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

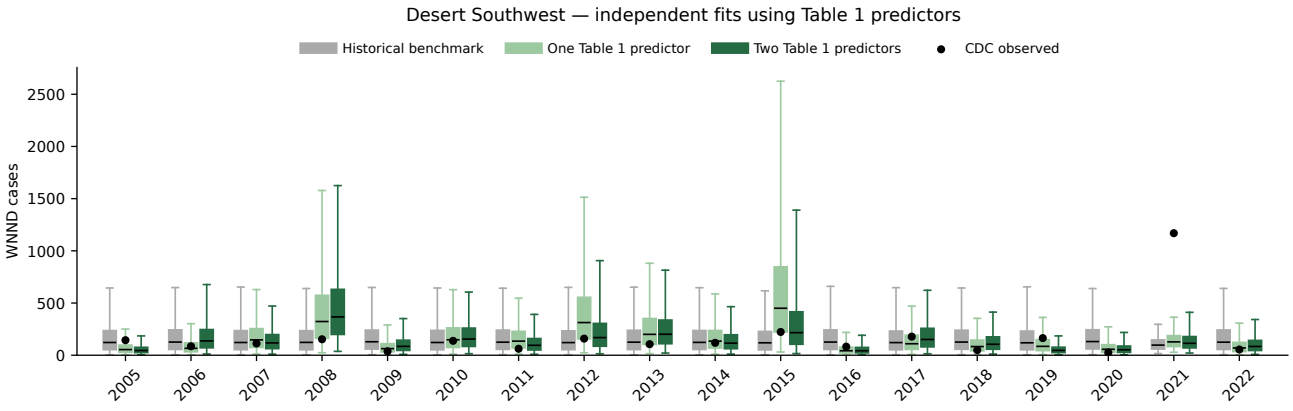


Figure R27: Figure S20: Desert Southwest annual probabilistic forecasts. independently reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

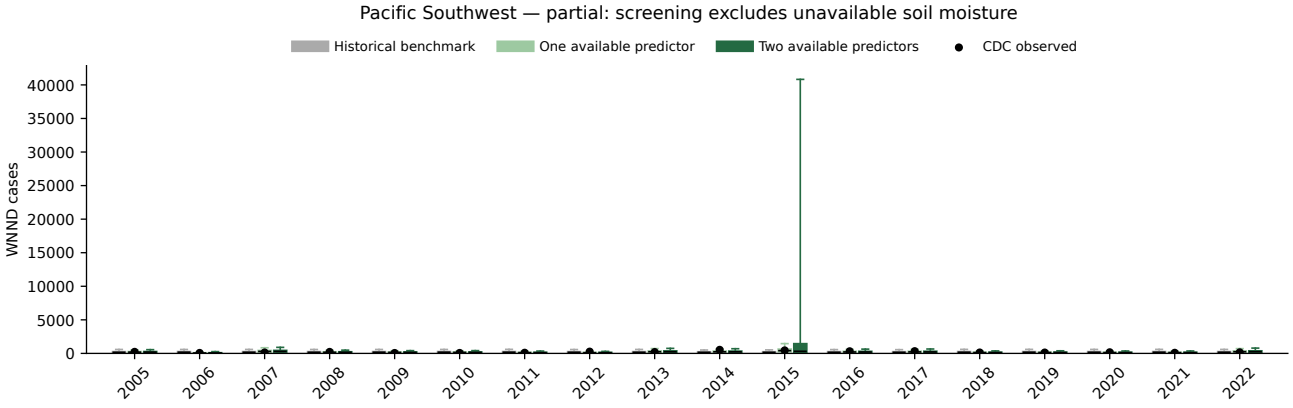


Figure R28: Figure S21: Pacific Southwest annual probabilistic forecasts. partially reproduced. Box/whisker definitions match Figure 4. Titles distinguish published predictor sets from restricted-candidate models. Black points are CDC observations; all forecasts are independently generated.

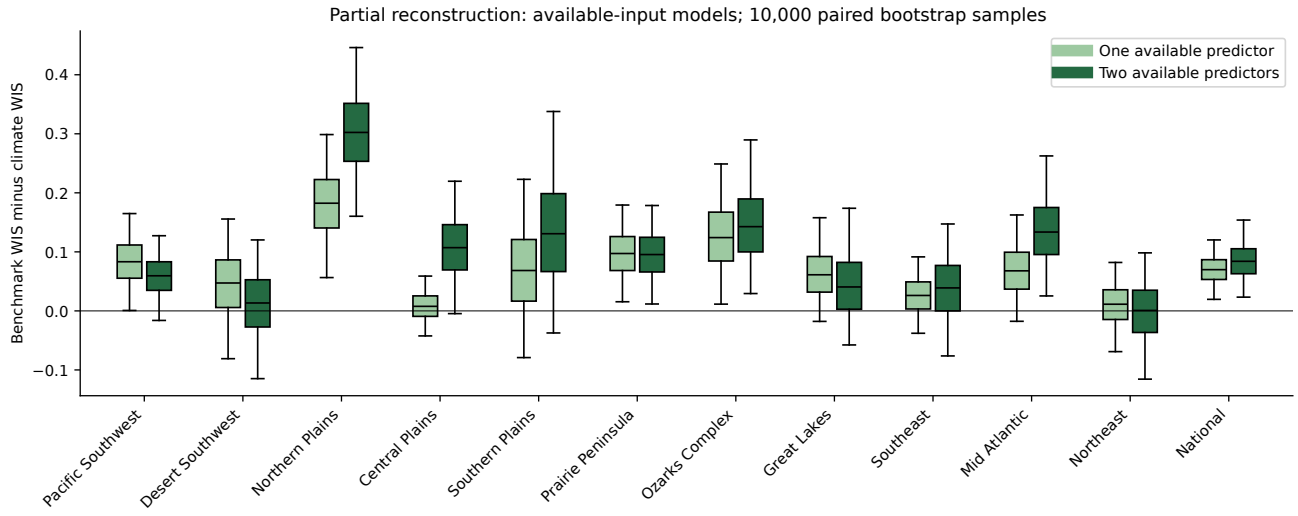


Figure R29: Figure S22: Absolute WIS improvement. partially reproduced. Absolute log1p WIS improvement from 10,000 paired bootstrap samples; positive values favor climate information. Green shades distinguish one and two predictors.

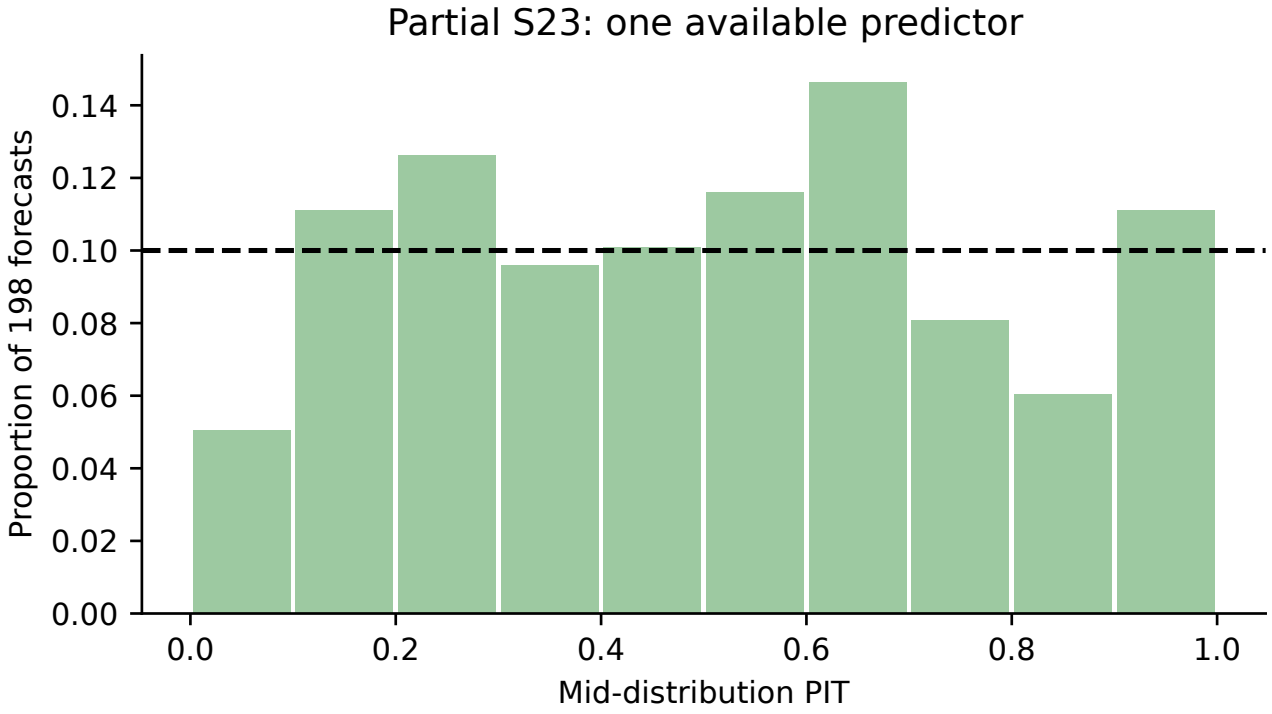


Figure R30: Figure S23: univariable PIT calibration histogram. partially reproduced. Proportion of 198 held-out observations in each of ten midpoint-PIT bins. The dashed line is 0.10; nonuniformity indicates calibration departures under the stated discrete convention.

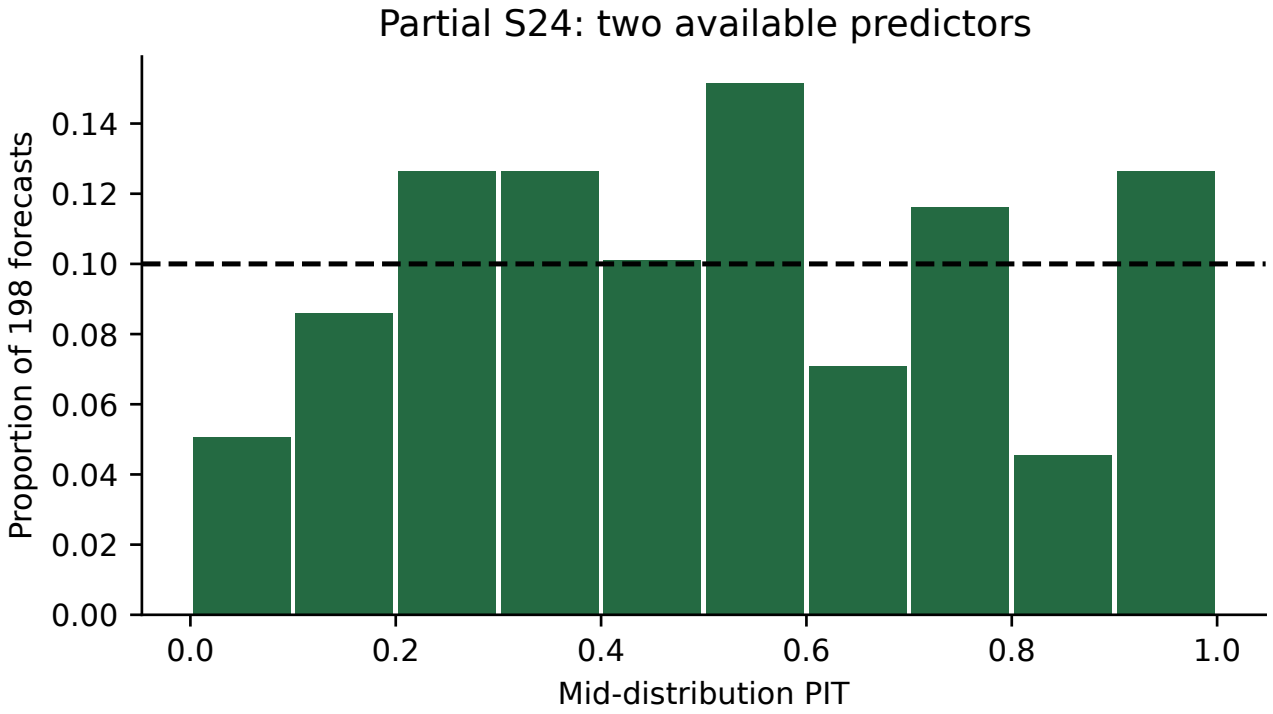


Figure R31: Figure S24: bivariable PIT calibration histogram. partially reproduced. Proportion of 198 held-out observations in each of ten midpoint-PIT bins. The dashed line is 0.10; nonuniformity indicates calibration departures under the stated discrete convention.

References

Harp, R. D., et al. (2026). A Regionally Determined Climate-Informed West Nile Virus Forecast Technique. *GeoHealth*, 10(2), e2025GH001657. https://doi.org/10.1029/2025GH001657. Supporting Information S1, Texts S1–S3, Figures S1–S24, Tables S1–S7.

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